

1: [10 pts.]

In order for a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ to be a norm it must satisfy the following properties:

1. $\|x\| \geq 0, \forall x \in \mathbb{R}^n$
2. $\|x\| = 0 \Leftrightarrow x = 0$
3. $\|cx\| = |c|\|x\|, \forall x \in \mathbb{R}^n, \forall c \in \mathbb{R}$
4. $\|x + y\| \leq \|x\| + \|y\|, \forall x, y \in \mathbb{R}^n$

For the function $\|x\|_1 = \sum_{i=1}^n |x_i|$ we have

1. The 1st property holds trivially.
2. Obviously, $x = 0 \Rightarrow \|x\|_1 = \sum_{i=1}^n |0| = 0$. Moreover, if $\|x\|_1 = 0 \Rightarrow |x_i| = 0, \forall i \Rightarrow x_i = 0, \forall i \Rightarrow x = 0$
3. $\|cx\|_1 = \sum_{i=1}^n |cx_i| = \sum_{i=1}^n |c||x_i| = |c| \sum_{i=1}^n |x_i| = |c|\|x\|_1$
4. $\|x + y\|_1 = \sum_{i=1}^n |x_i + y_i| \leq \sum_{i=1}^n (|x_i| + |y_i|) = \sum_{i=1}^n |x_i| + \sum_{i=1}^n |y_i| = \|x\|_1 + \|y\|_1$, where we used the triangle inequality.

Thus, $\|x\|_1$ is a norm.

For the function $\|x\|_\infty = \max_i |x_i|$ we have

1. The 1st property holds trivially.
2. Obviously, $x = 0 \Rightarrow \|x\|_\infty = \max_i |0| = 0$. Moreover, if $\|x\|_\infty = 0 \Rightarrow |x_i| = 0, \forall i \Rightarrow x_i = 0, \forall i \Rightarrow x = 0$
3. $\|cx\|_\infty = \max_i |cx_i| = \max_i |c||x_i| = |c| \max_i |x_i| = |c|\|x\|_\infty$
4. $\|x + y\|_\infty = \max_i |x_i + y_i| \leq \max_i \{|x_i| + |y_i|\} \leq \max_i \{|x_i|\} + \max_i \{|y_i|\} = \|x\|_\infty + \|y\|_\infty$, where we used the triangle inequality and some characteristic properties of the max function.

Thus, $\|x\|_\infty$ is a norm.

2: [5 pts.]

Utilizing the orthogonality of vectors x, y , that is $x^T y = \langle x, y \rangle = 0$ and the fact that $\langle x, x \rangle = \|x\|_2^2$ we get

$$\begin{aligned}\|x + y\|_2^2 &= (x + y)^T(x + y) = x^T x + x^T y + y^T x + y^T y \\ &= \|x\|_2^2 + 2x^T y + \|y\|_2^2 = \|x\|_2^2 + \|y\|_2^2\end{aligned}$$

3: [5pts]

$$\langle x, z \rangle = \left\langle x, y - \frac{\langle x, y \rangle}{\langle x, x \rangle} x \right\rangle = \langle x, y \rangle - \frac{\langle x, y \rangle}{\langle x, x \rangle} \langle x, x \rangle = 0$$

4: [10 pts.]

First of all, notice that for $x = 0$ or $y = 0$ the inequality $\langle x, y \rangle \leq \|x\|_2 \|y\|_2$ holds trivially. So, we prove the inequality for the rest of the cases.

$$\begin{aligned}\|y\|_2^2 &= \|z + \frac{\langle x, y \rangle}{\langle x, x \rangle} x\|_2^2 = \|z\|_2^2 + \left\| \frac{\langle x, y \rangle}{\|x\|_2^2} x \right\|_2^2 = \|z\|_2^2 + \left(\frac{\langle x, y \rangle}{\|x\|_2^2} \right)^2 \|x\|_2^2 = \|z\|_2^2 + \frac{(\langle x, y \rangle)^2}{\|x\|_2^2} \geq \frac{(\langle x, y \rangle)^2}{\|x\|_2^2} \\ \Rightarrow \|y\|_2^2 &\geq \frac{(\langle x, y \rangle)^2}{\|x\|_2^2} \Rightarrow \|x\|_2^2 \|y\|_2^2 \geq (\langle x, y \rangle)^2 \Rightarrow \langle x, y \rangle \leq \|x\|_2 \|y\|_2\end{aligned}$$

5: [15 pts.]

We can calculate the gradient either by using the chain rule for vector-valued functions (Bertsekas, App. A.5) or calculating the 1st order partial derivatives $\frac{\partial f}{\partial x_i}$ and formulating the respective vector (and if possible expressing it in a compact form). Note that I use both ways below and in addition I follow the convention that the gradient is a column vector.

$$1) f(\mathbf{x}) = \frac{1}{2} \|\mathbf{x} - \mathbf{b}\|^2 = \frac{1}{2} \sum_{i=1}^n (x_i - b_i)^2, \quad \mathbf{x} \in \mathbb{R}^n, \mathbf{b} \in \mathbb{R}^n$$

- The partial derivative of f w.r.t x_i is $\frac{\partial f}{\partial x_i} = \frac{1}{2} \cdot 2(x_i - b_i) = x_i - b_i$. Thus, the gradient of f is $\nabla f(x) = [\frac{\partial f}{\partial x_1}, \dots, \frac{\partial f}{\partial x_n}]^T = [x_1 - b_1, \dots, x_n - b_n] = \mathbf{x} - \mathbf{b}$.
- The 2nd order partial derivative of f w.r.t x_i and x_j is $\frac{\partial^2 f}{\partial x_i \partial x_j} = 1$ if $i = j$ and 0 if $i \neq j$. As a result, the Hessian of f is $\nabla^2 f(\mathbf{x}) = I$ (identity matrix).

$$2) f(\mathbf{x}) = \frac{1}{2} \|\mathbf{a}^T \mathbf{x} - b\|^2, \quad \mathbf{x} \in \mathbb{R}^n, \mathbf{a} \in \mathbb{R}^n, b \in \mathbb{R}$$

- Let $h(\mathbf{x}) = \frac{1}{2} \|\mathbf{x}\|_2^2$ and $g(\mathbf{x}) = \mathbf{a}^T \mathbf{x} - b$. Then, we can easily see that $\nabla h(\mathbf{x}) = \mathbf{x}$ and $\nabla g(\mathbf{x}) = \mathbf{a}$. Thus, using the chain rule, the gradient of f is $\nabla f(\mathbf{x}) = \nabla g(\mathbf{x}) \nabla h(g(\mathbf{x})) = \mathbf{a}(\mathbf{a}^T \mathbf{x} - b)$.

- From the result above notice that $\frac{\partial f}{\partial x_i} = a_i(\mathbf{a}^T \mathbf{x} - b)$. Thus, computing the 2nd order derivative w.r.t x_i and x_j we get $\frac{\partial^2 f}{\partial x_i \partial x_j} = a_i a_j$. Finally, notice that the Hessian can be written compactly as $\nabla^2 f(\mathbf{x}) = \mathbf{a}\mathbf{a}^T$ (notice that $\mathbf{a}\mathbf{a}^T$ is a matrix and is different from $\mathbf{a}^T \mathbf{a}$ which is a scalar).

3) $f(\mathbf{x}) = \exp\{\mathbf{b}^T \mathbf{x} + c\}$, $\mathbf{x} \in \mathbb{R}^n, \mathbf{b} \in \mathbb{R}^n, c \in \mathbb{R}$

- Let $h(x) = \exp(x)$ (x here is a scalar) and $g(\mathbf{x}) = \mathbf{b}^T \mathbf{x} + c$. Then, we can see that $\nabla h(x) = \exp(x)$ and $\nabla g(\mathbf{x}) = \mathbf{b}$. Then, using the chain rule the gradient of f is $\nabla f(\mathbf{x}) = \nabla g(\mathbf{x}) \nabla h(g(\mathbf{x})) = \mathbf{b} \exp\{\mathbf{b}^T \mathbf{x} + c\}$.
- We have that $\frac{\partial f}{\partial x_i} = b_i \exp\{\mathbf{b}^T \mathbf{x} + c\}$, $\frac{\partial^2 f}{\partial x_i \partial x_j} = b_i b_j \exp\{\mathbf{b}^T \mathbf{x} + c\}$. Using the same reasoning as in the previous examples we conclude that the Hessian of f is $\nabla^2 f(\mathbf{x}) = \mathbf{b}\mathbf{b}^T \exp\{\mathbf{b}^T \mathbf{x} + c\}$.

6: [15 pts.]

a)

For some arbitrary x, y we define the function $g(z) = f(z) - \frac{f(x)-f(y)}{x-y}z$, $x \neq y$. Notice that $g(z)$ is continuous in $[x, y]$, differentiable in (x, y) and it holds that $g(x) = g(y)$. Also, the derivative of g is $g'(z) = f'(z) - \frac{f(x)-f(y)}{x-y}$. Therefore, the conditions of Rolle's theorem are satisfied and so there exists a $x^{mid} \in (x, y)$ such that

$$g'(x^{mid}) = 0 \Rightarrow f'(x^{mid}) - \frac{f(x) - f(y)}{x - y} = 0$$

$$f(x) - f(y) = f'(x^{mid})(x - y)$$

b)

For some arbitrary x, y we define the function $h(z) = f(z) - f(y) - f'(y)(z - y) - c(z - y)^2$, $x \neq y$, where $c = \frac{f(x)-f(y)-f'(y)(x-y)}{(x-y)^2}$. Notice that $h(z)$ is continuous in $[x, y]$, differentiable in (x, y) and it holds that $h(x) = h(y) = 0$. Also, the derivative of h is $h'(z) = f'(z) - f'(y) - 2c(z - y)$. Therefore, the conditions of Rolle's theorem are satisfied and so there exists a $z^{mid} \in (x, y)$ such that $h'(z^{mid}) = 0$.

Moreover, notice that h' is continuous in $[z^{mid}, y]$, differentiable in (z^{mid}, y) , $h'(y) = 0$ and $h''(z) = f''(z) - 2c$. Therefore, utilizing Rolle's theorem we obtain a $x^{mid} \in (z^{mid}, y) \subset (x, y)$ such that

$$h''(x^{mid}) = f''(x^{mid}) - 2\frac{f(x) - f(y) - f'(y)(x - y)}{(x - y)^2} = 0 \Rightarrow$$

$$f(x) - f(y) = f'(y)(x - y) + \frac{1}{2}f''(x^{mid})(x - y)^2$$

7: Exercise 1.1.1 in the textbook [15 pts.]

Apply first order optimality condition, setting gradient to 0, we have

$$2x + \beta y + 1 = 0 \quad (1)$$

$$\beta x + 2y + 2 = 0 \quad (2)$$

It can be readily seen that when $\beta = \pm 2$, the above equations are contradictory and do not have solutions. In these cases, there is no stationary points of $f(x, y)$.

When $\beta \neq \pm 2$, solving (1) and (2), we have

$$x = \frac{2 - 2\beta}{\beta^2 - 4} \quad (3)$$

$$y = \frac{4 - \beta}{\beta^2 - 4} \quad (4)$$

Now we examine the second order optimality condition, the Hessian of $f(x, y)$ is

$$\begin{bmatrix} 2 & \beta \\ \beta & 2 \end{bmatrix}$$

Its eigenvalues are $2 + \beta$ and $2 - \beta$. The Hessian is positive semidefinite and $f(x, y)$ is convex when $-2 \leq \beta \leq 2$.

Summarizing all above with optimality conditions, we have following cases.

1. When $-2 < \beta < 2$, the Hessian is positive definite and $f(x, y)$ is convex, the stationary point in (3)-(4) is a global minimum.
2. When $\beta > 2$ or $\beta < -2$, the Hessian has both positive and negative eigenvalues, the stationary point in (3)-(4) is a saddle point.
3. When $\beta = \pm 2$, there is no stationary points.

8: Exercise 1.1.2 in the textbook [25 pts]

(a) Setting $\nabla f(x, y)$ to 0, we have

$$4x(x^2 - 4) = 0 \quad (5)$$

$$2y = 0 \quad (6)$$

The solution is

$$x = 0, \pm 2 \quad (7)$$

$$y = 0 \quad (8)$$

Hence, $f(x, y)$ has 3 stationary points

$$(x, y) = \begin{cases} (0, 0) \\ (-2, 0) \\ (2, 0) \end{cases}$$

The Hessian of $f(x, y)$ is

$$\begin{bmatrix} 12x^2 - 16 & 0 \\ 0 & 2 \end{bmatrix}$$

It can be easily seen that points (2,0) and (-2,0) have positive definite Hessian and Hessian at point (0,0) is neither positive semidefinite nor negative semidefinite.

From second order optimality conditions, we know (2,0) and (-2,0) are local optima, (0,0) is neither a local maximum nor a local minimum.

Substitute (2,0) and (-2,0) to $f(x, y)$,

$$f(2, 0) = f(-2, 0) = 0.$$

Since $f(x, y)$ is sum of squares, we have $f(x, y) \geq 0$.

Thus, (2,0) and (-2,0) are global optima.

(b)

Setting $\nabla f(x, y)$ to 0, we have

$$x + \cos y = 0 \tag{9}$$

$$-x \sin y = 0 \tag{10}$$

Solutions of (9)-(10) are

$$y = k\pi \tag{11}$$

$$x = -\cos k\pi \tag{12}$$

or

$$y = k\pi + \frac{1}{2}\pi \tag{13}$$

$$x = 0 \tag{14}$$

where k is any integer

The Hessian of $f(x, y)$ is

$$\begin{bmatrix} 1 & -\sin y \\ -\sin y & -x \cos y \end{bmatrix}$$

Eigenvalues of $\nabla^2 f(x, y)$ are 1 and $-x \cos y - \sin^2 y$

For points with form (11)-(12), we have both $\nabla f(x, y) = 0$ and $\nabla^2 f(x, y) > 0$, thus they are local minima

For points with form (13)-(14), their Hessian have both positive and negative eigenvalues, they are saddle points.

Therefore, all local optima are

$$y = k\pi \quad (15)$$

$$x = -\cos k\pi \quad (16)$$

where k is any integer.

(c)

Setting $\nabla f(x, y)$ to 0, we have

$$\cos x + \cos(x + y) = 0 \quad (17)$$

$$\cos y + \cos(x + y) = 0 \quad (18)$$

Solving equations above, we get 3 points

$$(x, y) = \begin{cases} (\frac{\pi}{3}, \frac{\pi}{3}) \\ (\pi, \pi) \\ (\frac{5}{3}\pi, \frac{5}{3}\pi) \end{cases}$$

The Hessian is

$$\begin{bmatrix} -\sin x - \sin(x + y) & -\sin(x + y) \\ -\sin(x + y) & -\sin y - \sin(x + y) \end{bmatrix}$$

The eigenvalues are $-\sin x - \sin(x + y)$ and $-\sin y - \sin(x + y) + \frac{\sin^2(x+y)}{-\sin y - \sin(x+y)}$ when $-\sin x - \sin(x + y) \neq 0$

Substitute $(\frac{\pi}{3}, \frac{\pi}{3})$ and $(\frac{5}{3}\pi, \frac{5}{3}\pi)$ to the eigenvalues above, we know the Hessian is negative definite at $(\frac{\pi}{3}, \frac{\pi}{3})$ and positive definite at $(\frac{5}{3}\pi, \frac{5}{3}\pi)$.

Thus, $(\frac{\pi}{3}, \frac{\pi}{3})$ is a local maximum and $(\frac{5}{3}\pi, \frac{5}{3}\pi)$ is a local minimum.

Substitute (π, π) to Hessian, the Hessian is 0, but this point is neither a local minimum nor a local maximum since $f(x, y)$ will decrease in direction(1,1) and increase in direction (-1,-1).

Summarizing all above,

$$(\frac{\pi}{3}, \frac{\pi}{3}) \text{ local maximum} \quad (19)$$

$$(\frac{5}{3}\pi, \frac{5}{3}\pi) \text{ local minimum} \quad (20)$$

(d)

Setting $\nabla f(x, y)$ to 0, we have

$$-4x(y - x^2) - 2x = 0 \quad (21)$$

$$2(y - x^2) = 0 \quad (22)$$

The only solution to above equations is

$$x = 0 \quad (23)$$

$$y = 0 \quad (24)$$

The Hessian of $f(x, y)$ is

$$\begin{bmatrix} -4y + 12x^2 - 2 & -4x \\ -4x & 2 \end{bmatrix}$$

At point (0,0) the Hessian matrix is neither positive semidefinite nor negative semidefinite, thus (0,0) is neither a local minimum nor a local maximum.